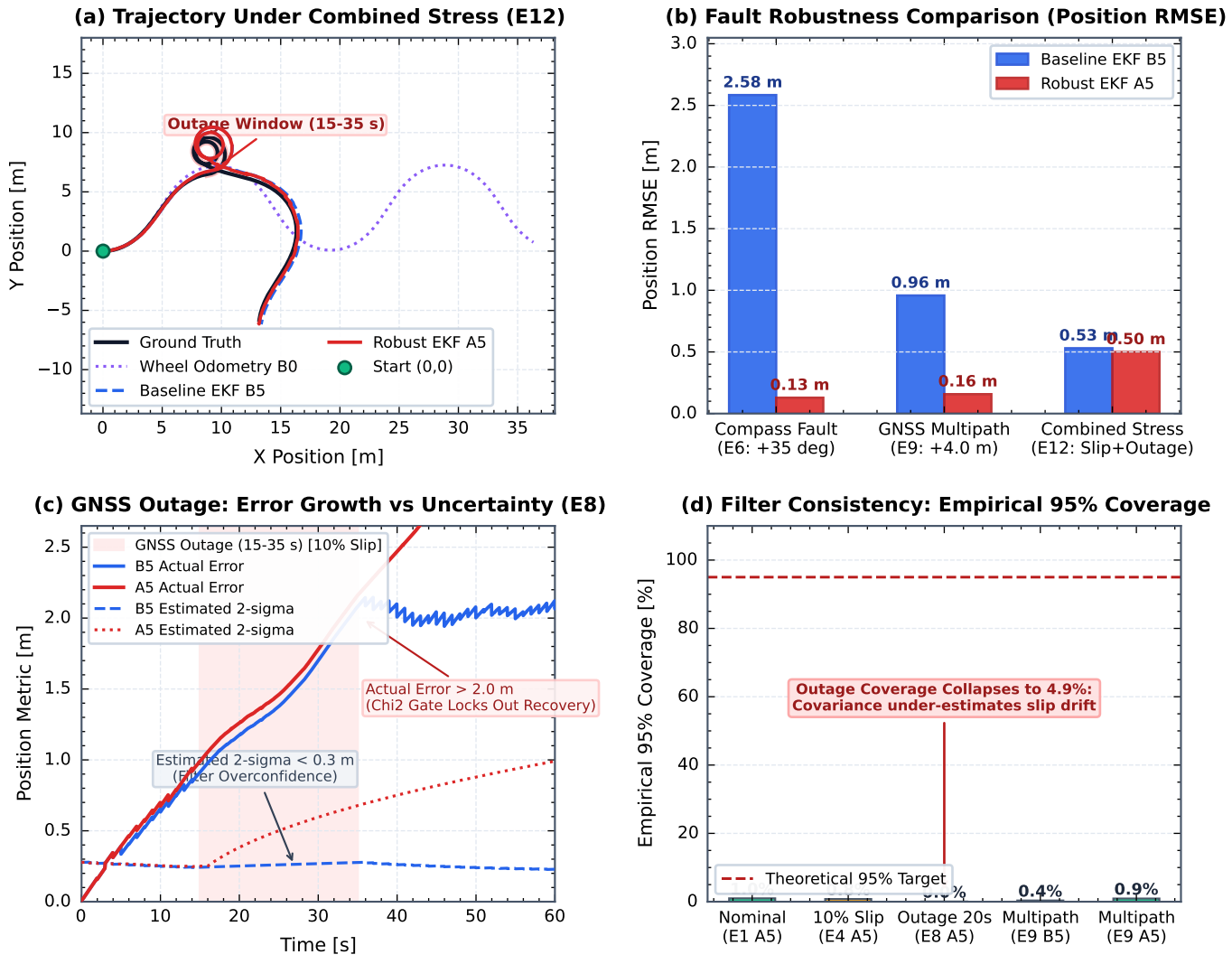

(a) Trajectory Under Combined Stress (E12)
Outage Window (15-35 s)
Ground Truth
Wheel Odometry B0
Baseline EKF B5
Robust EKF A5
Start (0,0)
X Position [m]
Y Position [m]
(b) Fault Robustness Comparison (Position RMSE)
Baseline EKF B5
Robust EKF A5
2.58 m
0.13 m
0.96 m
0.16 m
0.53 m
0.50 m
Position RMSE [m]
Compass Fault
(E6: +35 deg)
GNSS Multipath
(E9: +4.0 m)
Combined Stress
(E12: Slip+Outage)
(c) GNSS Outage: Error Growth vs Uncertainty (E8)
GNSS Outage (15-35 s) [10% Slip]
B5 Actual Error
A5 Actual Error
B5 Estimated 2-sigma
A5 Estimated 2-sigma
Actual Error > 2.0 m
(Chi2 Gate Locks Out Recovery)
Estimated 2-sigma < 0.3 m
(Filter Overconfidence)
Position Metric [m]
Time [s]
(d) Filter Consistency: Empirical 95% Coverage
Outage Coverage Collapses to 4.9%:
Covariance under-estimates slip drift
Theoretical 95% Target
0.4%
0.9%
Empirical 95% Coverage [%]
Nominal
(E1 A5)
10% Slip
(E4 A5)
Outage 20s
(E8 A5)
Multipath
(E9 B5)
Multipath
(E9 A5)